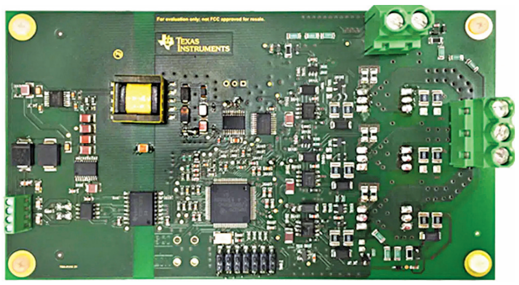
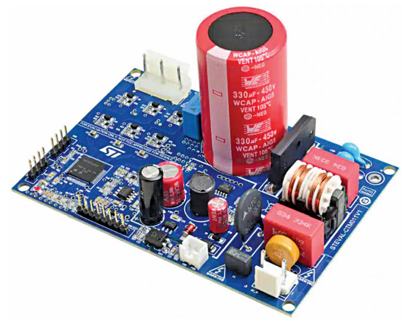
Interesting Reference Designs OF COMPRESSORS

Title of Reference Design	HVAC Compressor	250-Watt Mainstream Compressor
Image		
Key Attractions	The reference design for the HVAC compressor excels in heavy load startups, low-speed precision, and robust safety features, optimising electric vehicle battery efficiency and performance within stringent automotive standards.	The 250-watt compressor reference design features a three-phase inverter at its heart. The solution, enhanced by a 2-shunt version, supports advanced motor control and offers a reduced PCB footprint for diverse industrial applications.
Highlights	The TIDA-01418, a reference design from Texas Instruments (TI) is a three-phase inverter drive employing discrete IGBTs to power BLDC motors with a sensorless torque control mechanism. The system utilises the UCC27712-Q1 high- and low-side gate driver alongside discrete IGBT half bridges. At the heart of the system is TI's C2000 Piccolo F2805x microcontroller, integrated with the InstaSPIN-FOC sensorless motor control solution. This advanced setup enables motor parameter identification and automatic tuning of the current loop, ensuring high-performance sensorless functionality. The design offers key features like starting under heavy loads and precise low-speed control that enhances the dynamic range of compressors, enabling low-speed operation with full torque, aiding in electric vehicle battery efficiency during HVAC operation. The design handles low-voltage conditions up to 45V and withstands reverse battery and load dump scenarios in 12V systems. It includes safety features such as an active Miller clamp, overcurrent detection circuit, isolated CAN-communication interface for secure data transmission, non-isolated gate drivers with a 2.5A peak output, and an integrated isolated power supply, all within a compact layout that adheres to automotive temperature and quality standards.	The STEVAL-CTM011V1 from STMicroelectronics is a 250-watt compressor reference design focused on a three-phase inverter, powered by the integrated STSPIN32F0601Q controller. This controller combines a 3-phase 600-volt gate driver with an Arm Cortex-M0 STM32 microcontroller unit (MCU), and the design includes a STGD5H60DF insulated-gate bipolar transistor (IGBT). The design uses a system-in-package approach, offering an energy-efficient solution with integrated protection. The performance is enhanced in its 2-shunt version, providing an advantage over other market alternatives. The system's power supply stage includes a VIPER122 high-voltage converter in a buck configuration, generating supply voltages for the compressor's operation and supports both one-shunt and 2+1 shunt sensing topologies. The board is equipped with STSW-CTM011 reference firmware, facilitating the implementation of sensorless field-oriented control (FOC) and driving permanent magnet synchronous motors (PMSMs) and BLDC motors in various industrial appliances. The design has low standby power consumption, overcurrent and undervoltage protections, and an inverter power stage rated at 600V and 5A. It can drive motors up to 250W and features a reduced PCB footprint and a minimised BoM, leading to a more efficient and compact design.
Applications	The reference design is suitable for driving BLDC motors in automotive air-conditioning systems.	The reference design is suitable for variable-speed 3-phase BLDC compressor motors used in refrigeration systems, air compressors, and AC units.
OEM Brand	Texas Instruments (TI)	STMicroelectronics
URL	https://www.electronicsforu.com/electronics-projects/reference-design-for-hvac-compressor	https://www.electronicsforu.com/electronics-projects/250-watt-mainstream-compressor-reference-design